INTRODUCTION TO RESEARCH

- It is an Art of Scientific Investigation.
- According to Redman and Mory, Research is a “Systematized effort to gain new knowledge”.
- Research is an original addition to the available knowledge, which contributes to its further advancement.
- In sum, Research is the search for knowledge, using objective and systematic methods to find solution to a problem.

Research Methods vs. Methodology

Research Methods are the methods that the researcher adopts for conducting the research studies.

Research Methodology is the way in which research problems are solved systematically.

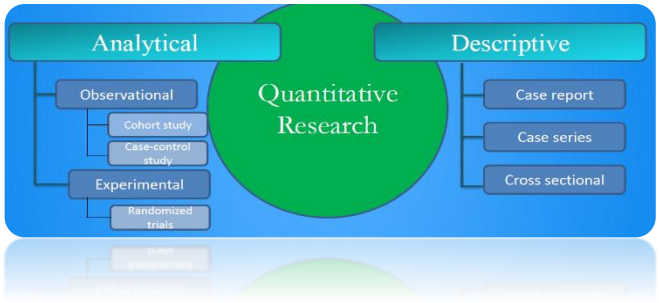
It is the Science of studying how research is conducted Scientifically.

Significance of Research

- Research inculcates scientific and inductive thinking and it promotes the development of logical habits of thinking and organization.

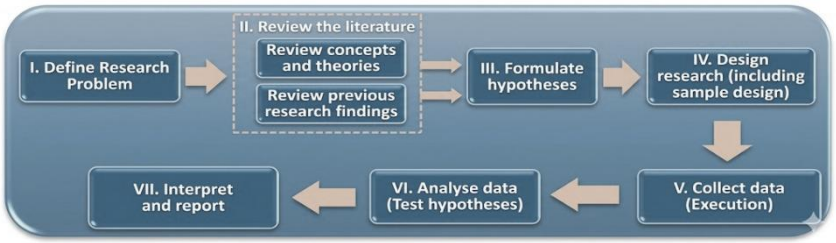
Types of Research

Quantitative	Qualitative	Mixed
• numbers, numbers, numbers. • quantitative methods use data that can be counted or converted into numerical form to address "How?" questions.	• words, words, words. • Qualitative research refers to the use of non-numerical observations to answer "Why?" questions	



Basic Question in Research Are exposure and disease/outcome linked? Is there an association between them? **Exposure => Disease / Health Outcome**

Research Process



I. Defining the Research Problem

- A research problem, in general, refers to some difficulty which a researcher experiences in the context of either a theoretical or practical situation and wants to obtain a solution for the same.
- The research problem undertaken for study must be carefully selected. Help may be taken from a research guide in this connection.

The following points may be observed by a researcher in selecting a research problem or a subject for research:

1. **Subject which is overdone** should not be normally chosen, for it will be a difficult task to throw any new light in such a case.
2. There must be **some objective(s)** to be attained at. If one wants nothing, one cannot have a problem.
3. The subject selected for research should be **familiar and feasible** so that the related research material or sources of research are within one's reach.
4. The **importance** of the subject, the **qualifications** and the **training of a researcher**, the **costs i**nvolved**, the time factor** are few other criteria that must also be considered in selecting a problem. Before the final selection of a problem is done, a researcher must ask himself the following questions:

   I. Whether he is well equipped in terms of his background to carry out the research?
   II. Whether the study falls within the budget he can afford?
   III. Whether the necessary cooperation can be obtained from those who must participate in research as subjects?

5. If the field of inquiry is **relatively new and does not have available** set of well-developed techniques, a brief feasibility study must always be undertaken.

- Defining a research problem properly and clearly is a crucial part of a research study and must in no case be accomplished hurriedly.
- The technique for the purpose involves the undertaking of the following steps generally one after the other:

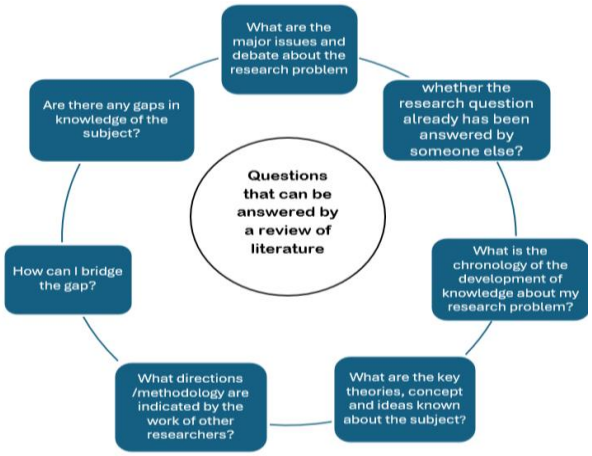
  IV. Statement of the problem in a general way;
  V. understanding the nature of the problem;
  VI. surveying the available literature
  VII. developing the ideas through discussions; and
  VIII. rephrasing the research problem into a working proposition.

II. Literature Review

Once the problem is formulated, the researcher should undertake extensive literature review connected with the problem.

Why Literature Review??????

a) Assist in refining statement of the problem.
b) Strengthening the argument of selection of a research topic (Justification).
c) it helps to get familiar with various types of methodology that might be used in the study (Design).



Sources of Literature:

Books
- TEXT BOOKS
- MONOGRAPHS
- EDITED COLLECTIONS

Vital statistics
- Census
- Government Records
- Surveillance system
- Surveys

Journal Articles
- Academic journals
- Conference Proceedings

International organization documents
- e.g. (WHO, UNICEF)

Indexing and Abstracting journal search engines
- PubMed
- Google Scholar

Media
- Newspaper
- Magazine

Past Dissertations

Internet

- ***Finding too much?*** If you find so many citations that there is no end in sight to the number of references you could use, it's time to re-evaluate your question. It's too broad/Nothing much to explore
- ***Finding too little?*** On the other hand, if you can't find much of anything, ask yourself if you're looking in the right area.
- **Take thorough notes**. Be sure to write copious notes on everything as you proceed through your research. It's very frustrating when you can't find a reference found earlier that now you want to read in full.
- Look for references to papers from which you can identify the most useful journals.
- Identify those authors who seem to be important in your subject area.
- **Institutional library** serves as a greatest source of literature review.
- Talk to the librarian for greater insight on the number of journals available either as a hard copy or online subscription

- Our JNMC library subscribes 115 International/Foreign and 25 Indian Journals in various specialties. The library has a exclusive collection of about 2000 Thesis and Dissertations of MD/MS/PhD students besides a comprehensive collection of WHO Publications.
- Besides this it also provides access to various consortia e.g. ERMED (2000 Journals),J-Gate, UGC Info-net, Pub Med database of 18 million references/documents and other open-source documents.

Important concept related to academic journals

Indexing- Indexing as defined by British indexing standard (BS3700:1988), as a systematic arrangement of entries designed to enable users to locate information in a document.

- Many commercial indexing services available.
- Quality indexing services includes PubMed, Scopus, Embase etc
  - A good indexing bodies ensures that journal should have Content, which is of high-quality.
  - It should follow peer-review process.
  - Subject matter of the journal should be compatible with the scope of Indexing body.
  - Disciplined publishing history.
- Now a days predatory publishers' and predatory journals' brag about how many abstracting and indexing services cover their journals. (Check... may b they r lying!!!!!!!!!)

III. Development of Working Hypothesis

- After extensive literature survey, researcher should state in clear terms the working hypothesis.
- For a researcher hypothesis is *a formal question that he intends to resolve*.
- A hypothesis is a **proposed explanation for an observable phenomenon which is capable of being tested by scientific methods**.
- For example, consider a statement:

"The drug A is equally efficacious as drug B."

- This is a hypothesis capable of being objectively verified and tested.

Characteristics of hypothesis: Hypothesis must possess the following characteristics:

- **Hypothesis should be clear and precise. If the hypothesis is not clear and precise, the inferences drawn on its basis cannot be taken as reliable.**
- **Hypothesis should be capable of being tested.**
- **Hypothesis should be limited in scope and must be specific.**
- **Hypothesis should be stated as far as possible in most simple terms so that the same is easily understandable by all concerned.**
- **Hypothesis should be amenable to testing within a reasonable time. One should not use even an excellent hypothesis, if the same cannot be tested in reasonable time for one cannot spend a life-time collecting data to test it.**
- **Thus, hypothesis must actually explain what it claims to explain.**

IV. Research Design

"A research design is the arrangement of conditions for collection and analysis of data in a manner that aims to combine relevance to the research purpose with economy in procedure." Research Methods in Social Sciences, 1962, p. 50

- It constitutes the blueprint for the collection, measurement and analysis of data.
- An outline of what the researcher will do from writing the hypothesis and its operational implications to the final analysis of data.

Important concepts relating to research design:

1. Dependent and independent variables:

- A concept which can take on different quantitative values is called a ***variable***. As such the concepts like weight, height are all examples of variables.
- Phenomena which can take on quantitatively different values even in decimal points are called '**continuous variables**.
- If it can only be expressed in integer values, they are non-continuous variables or in statistical language '**discrete variables**.
- If one variable depends upon or is a consequence of the other variable, it is termed as a dependent variable, and the variable that is antecedent to the dependent variable is termed as an independent variable.
- For instance, *if we say that height depends upon age, then height is a dependent variable and age is an independent variable*.

2. Extraneous variable:

- Independent variables that are not related to the purpose of the study, but may affect the dependent variable are termed as extraneous variables or confounding variables.
- Whatever effect is noticed on dependent variable as a result of extraneous variable(s) is technically described as an '**experimental error**'.
- A study must always be so designed that the effect upon the dependent variable is attributed entirely to the independent variable(s), and not to some extraneous variable or variables.

3. Control:

- One important characteristic of a good research design is to minimize the influence or effect of extraneous variable(s).
- The technical term '**control**' is used when we design the study minimizing the effects of extraneous independent variables.
- In experimental researches, the term 'control' is used to refer to restrain experimental conditions.

4. **Experimental and control groups:**

- In experimental hypothesis-testing research when a group is exposed to usual conditions, it is termed a ***'control group'***, but when the group is exposed to some novel or special condition, it is termed an ***'experimental group'***

5. **Treatments**:

- The different conditions under which experimental and control groups are put are usually referred to as ***'treatments.***

Different Research Designs

Different research designs can be conveniently described as:

- Exploratory Research Design
- Descriptive and Diagnostic Research Design
- Hypothesis-testing Research Design/Experimental Research Design